

Dengue epidemic-forest paper: a few decisions for us

Nasution positioning, harmonised Strength of Linkage, and scope

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Hi Kamal, Nuning, and Maria,

I wanted to put down, in one place, the four things about the South Cotabato dengue epidemic-forest paper that I'd most like us to agree on early, while they're still easy to change. They are: (a) how I've positioned this paper relative to our group's Nasution 2024 Jakarta paper; (b) that I've **harmonised the Strength of Linkage to Nasution's exact Eq. 5**, which (c) **changed the paper's headline finding**; and (d) a **scope/title choice** now that I've applied the method across several Region 12 provinces. I've put a short list of the limitations I actually ran into at the end.

What I'm asking, in one line

Nuning and Kamal are on **both** this paper and the Nasution 2024 Jakarta paper, and Maria is a co-author here — so how we cite and build on the Jakarta work, and how boldly we frame the multi-province results, feel like decisions we should make together and up front. I think both moves *strengthen* the paper, but I want your read.

The sister paper

Nasution, Sitorus, Sukandar, Nuraini, Apri, Salama (2024), *Scientific Reports* 14:7619 — the region-scale epidemic forest applied to acute respiratory infection in Jakarta.

Here's what I think it already established, so I've attributed these to it and do **not** claim them as novel here:

- adapting Li et al.'s individual-level epidemic forest to **aggregated administrative units**;
- **case prevalence** as a third Strength-of-Linkage criterion (their Eq. 5);
- **Richards / logistic onset** estimation;
- R_t and R_c , dominant trees, and the intervention-by-case-suppression idea.

What's genuinely new here (and I think it's a real contribution)

Nasution restricts candidate parents to a **fixed binary adjacency** matrix and reports a **single deterministic forest** — and the Discussion there explicitly names “inter-regional connectivity and networks” as missing future work.

That's exactly what I've built here:

1. I replace the fixed adjacency with a **probabilistic distance-decay connectivity network** (Eq. 4, $P_{ij} = e^{-\lambda d}$);
2. I reconstruct the forest as a **Monte-Carlo ensemble** ($M = 5000$ realisations) with **per-edge stability** rather than one tree;
3. it's the **first Philippine dengue application** of the approach.

So the honest framing I've gone with is: prevalence, region-scale aggregation, **and now the exact Strength of Linkage (their Eq. 5)** are theirs; the **connectivity layer + ensemble + edge stability** are ours — we build the thing they flagged as future work. I think citing our Jakarta paper as the direct predecessor makes the lineage a strength, not an overlap.

I harmonised the Strength of Linkage — and it changed the headline result

The paper previously used an equivalent but re-parameterised SoL (normalised by the candidate mean, minimised), which I'd chosen for exact reproducibility. I've now **adopted Nasution's exact Eq. 5** (maximised, inverse min-normalised distances, prevalence difference), so **both papers use one identical linkage definition**. I think that's cleaner and it strengthens the “same method, new connectivity layer” story.

But it changed a central result, and Kamal, I'd really like your read before I lock it in:

- **Unchanged: General Santos City remains the consistent terminal recipient** (never seeds, all periods; $R_c = 0$ in 100% of the 5,000 realisations), and the global reproduction ratios are $R_t = 5 / 2 / 3$.
- **Changed: the dominant source flips from rural Surallah to Koronadal City — the provincial capital** (LGU reproduction number $R_c = 4$ in Periods 1–2; inland municipalities, Banga leading, take over in Period 3). The R_c : Koronadal City [4,4,0], Tupi [2,2,1], Surallah [2,1,2], Banga [0,0,2].
- **Narrative impact:** the framing moved from “rural periphery → urban centre” to “**inland administrative/population hub → coastal centre.**” I rewrote the prose to match the numbers, but the *interpretation* of why the capital seeds is a scientific judgement I'd rather we confirm together.

One thing that reassures me about the flip: an independent study (Ygoña, Deligero & Ceballos, 2025, *Recoletos Multidiscip. Res. J.*) analysed South Cotabato dengue for 2020–2022 with scan statistics (SaTScan / Moran's I) and found the **dominant spatio-temporal cluster over Polomolok and Koronadal City**, attributing the urban centre's high incidence to drainage and water-storage conditions. My ensemble reaches the **same conclusion (Koronadal City as leading source) by a different method on a non-overlapping window (2015–2019)** — so I read the harmonised result as corroborated, not a modelling artefact. I think that makes the Koronadal finding an easier sell.

How I've framed and positioned it

I've framed the paper as **methodological** — its contribution being *quantified linkage uncertainty* (the probabilistic connectivity network + Monte-Carlo ensemble + per-edge stability), demonstrated on the South Cotabato case, with references led by the epidemic-forest / transmission-reconstruction lineage and the outbreak literature supporting the application. I've grounded the

three study periods as the **2015, 2016–17, and 2019** outbreak seasons (Period 3 = the 2019 national epidemic).

The scope/title reframe — cross-province validation

Using the DOH Region 12 (CHD12) line-list, I applied the identical pipeline to **three further provinces — North Cotabato, Sultan Kudarat, and Sarangani** — as a generalisability check. The same signature recurs everywhere: an **inland population/administrative hub (the provincial capital in 3 of 4 provinces)** is the dominant source, with **peripheral/coastal LGUs terminal**. I went with a “**case study + preliminary validation**” framing rather than a full multi-province reframe: the **title** now reads “*...a South Cotabato case study with a preliminary cross-province transferability assessment in Region 12*” (I softened “validation” to “transferability”), and I updated the abstract/methods/discussion/conclusion to match, with a new results §3.5 and summary table. This lifts the generalisation into the **title/abstract**, so it’s a scope decision I’d like us to make together — the alternative is to keep the South Cotabato title and hold the reframe until the validation is final.

I’ve kept the validation **explicitly preliminary** and labelled it so throughout: the coordinates are gazetteer town-centres (not GIS centroids), the Sultan Kudarat/Sarangani populations are provisional (not PSA), and I reused the outbreak windows. Finalising it needs PSA populations, GIS centroids, and per-province period selection.

Limitations I actually hit (now a Limitations section in the paper)

I think we should be upfront about these, since I ran into several of them directly:

- **No ground truth / unsupervised** — many individual edges are weakly determined (stability < 40%), so I report the stable source/sink partition, not exact trees.
- **Aggregation** — links are between LGUs, not individuals; onset fits are shakier for low-count LGUs.
- **Model dependence** — the dominant source depends on the SoL weights/kernel (now probed by the fuller robustness suite in §3.4 / Table 8 — stable across weights, decay rates, kernel form, and a deterministic adjacency — but I still note it).
- **Surveillance data** — under-reporting and report-based onset dates.
- **(Resolved) Pipeline is now fully Python** — the whole analysis (smoothing, onset estimation, ensemble) runs natively in Python, so there is no longer any MATLAB dependency or cross-software onset caveat.
- **Preliminary metadata** for the validation provinces (as above).

A few specifics in the current draft worth flagging

So nothing catches you by surprise when you read the paper, a few choices I’ve made:

- **The cross-province work is framed as a *preliminary transferability assessment*, not a “validation”** — including in the title — since the metadata for those provinces is still provisional.
- **The out-degree metric is called the “LGU reproduction number”** (most of our units are municipalities, not cities). The symbol $R_{c,i}$ and the attribution to your Eq. 5 framework are unchanged.

- **The decay rate $\lambda = 5$ is justified explicitly** — it corresponds to a decay length of ≈ 22 km (half-probability ≈ 15 km), which matches the short-range dispersal of Aedes-borne dengue.
- **The Discussion explains why General Santos is always terminal** — it onsets after the inland hubs (so it reads as a recipient), being terminal is about timing not burden, and straight-line distance can't capture its long-range road/commuter links.
- **The three outbreak windows are justified** — chosen by visual inspection corroborated by DOH surveillance / the known epidemic years, with algorithmic change-point detection named as a future refinement.
- **There is a robustness suite (Table 8 + Figure 4).** I re-ran the full ensemble while varying λ over an order of magnitude, swapping the exponential kernel for a Gaussian one, and replacing the probabilistic network with a fixed **deterministic adjacency**. The result is reassuring: **General Santos stays terminal ($R_c = 0$) in every setting, and Koronadal (Periods 1–2) and Banga (Period 3) stay the dominant sources in all of them** — λ only changes the network density, and with it the R_t magnitude. The deterministic adjacency (essentially the Nasution-style fixed matrix) reproduces the same $R_t = 5/2/3$ but forces every edge to look certain — exactly the over-confidence our ensemble avoids.
- **It's fully reproducible** — a notebook section regenerates the robustness table and figure.

What I'd like us to agree on

- That we frame the paper as the **connectivity / ensemble follow-up** to Nasution 2024, citing it as the direct predecessor, with the attribution split (prevalence / region-scale + SoL = Nasution; network + ensemble + stability = here).
- **Sign-off on the harmonised SoL** (adopting their exact Eq. 5) and on the **resulting change in the headline finding** (Koronadal City, not Surallah, as the dominant source) — which Ygoña et al. (2025) independently corroborate. The old SoL is one toggle away (SOL_MODE="current") if we'd rather keep the Surallah result.
- The **methodological framing** (uncertainty quantification as the aim, South Cotabato as the demonstrating case).
- **The scope/title decision** — are we comfortable putting the preliminary cross-province transferability assessment in the **title and abstract**, or would you rather I keep it as the §3.5 subsection and hold the title until the metadata is final?
- **The R_c relabel** — I renamed “city reproduction number” to “**LGU reproduction number**” (most of our units are municipalities, not cities; the symbol $R_{c,i}$ and the attribution to your Eq. 5 framework are unchanged). Since the term comes from the Jakarta paper, tell me if you'd rather keep “city reproduction number” — it's a one-word revert.
- **Renamed both R_t and R_c** — to pre-empt a predictable reviewer objection (our quantities are not classical effective reproduction numbers), I renamed R_t to the “**LGU branching ratio (LBR)**” and R_c to the “**LGU out-degree index (LOI)**”, changing both names and symbols (R_t becomes LBR; R_c becomes LOI). These are shared with the Jakarta paper, so I'd like your read — both are clean find-and-replace reverts if the group prefers Nasution's original notation. This is shared terminology with the Jakarta paper, so I'd like your read — it's a clean find-and-replace to revert if you'd prefer we keep Nasution's

$$R_t$$

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- A sanity-check that the **limitations list** reads fairly — whether any should be softened or added.
- Whether there’s anything **in progress on your side** this should coordinate with (e.g. a planned extension of the Jakarta work).

Anchor references for context

- **Nasution et al. 2024**, *Sci Rep* 14:7619 — sister paper (DOI 10.1038/s41598-024-58390-3).
- **Li et al. 2019**, *Ann. AAG* 109(3):812–836 — original epidemic forest, individual-level, space + time only (DOI 10.1080/24694452.2018.1511413).
- **Haydon et al. 2003** — original epidemic *trees* at farm (unit) level; why LGU-scale reconstruction is a legitimate lineage, not a degraded Li.

Thanks for reading — happy to hop on a call if that’s easier.

Jayrold